DIELECTRIC MICROSPHERE-BASED OPTICAL SYSTEM FOR SUPER-RESOLUTION MICROSCOPY

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ABSTRACT

We present a dielectric microsphere-based super-resolution optical microscope that can image a sample beyond the classical diffraction limit [1]. The super-resolution imaging capability of a single microsphere normally is restricted to an area of ~10 μm^2, which essentially is the area of close contact of a microsphere with the sample. By fixing the microsphere on a frame attached to a microscope objective, we perform automatic step-by-step scanning and imaging of the sample. Later, the generated image set is synthesized into a single wide-field super-resolution image using a dedicated algorithm. Furthermore, we propose an array fabrication process for placing multiple microspheres in the optical path of the microscope objective, thereby parallelizing the imaging process.

KEYWORDS

Microsphere, super-resolution imaging, photonic nanojet, scanning optical microscopy.

INTRODUCTION

The resolution of a classical optical microscope is determined by Abbe's diffraction limit. A recently discovered way to overcome this limitation is converting sub-diffractional information close to the sample, which is carried by evanescent waves, into a propagating wave traveling to the far-field [2], [3]. Many methods were introduced where a micro- or nano-lens (*e.g.* nanoscale lens, polymer microdroplet or dielectric microsphere) was placed on the top of the sample, thus realizing transmission of the near-field information into the far-field [4]–[10]. Such methods are called super-resolution microscopies, as sample features with a size of λ/3-λ/7 can be resolved in the virtual image plane, where λ is the wavelength of the illumination [9], [11]. The mechanism behind these imaging techniques is based on the formation of a photonic nanojet, which is a narrow and strongly focused beam of light created by the micro- or nano-lens at the side where the light waves exit from it. Its shape and dimension is determined by the material and geometrical parameters of the lens. Typically, a nanojet is ~2λ long along the light path and has a width of ~λ/3 (full width at half maximum) [1]. The imaging mechanism of a dielectric microsphere was discussed in reference [1]. When the focused light that exits a microsphere exposes a sample to be imaged, no nanojet is generated, but instead the reflected light follows a reflection-symmetric optical path. At the same time, evanescent waves that contain the high spatial-frequency information of the object are converted into propagating waves within the microsphere. The imaging capability of the microsphere directly correlates with the geometrical properties of the focused photonic nanojet, as the same optical paths are involved. In imaging mode, even sub-diffraction features of the sample can be observed by a classical optical microscope (*Figure 1*). The main factor that stands in the way of widespread usage of these systems, is the limitation of the field of view, which is determined by the diameter of the lens used for imaging.

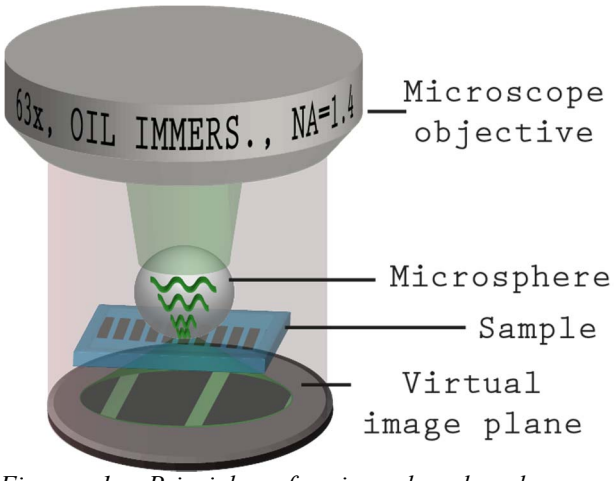

Figure 1: Principle of microsphere-based super-resolution imaging. A dielectric microsphere is placed directly on the sample, generating a virtual image that can be observed by a standard optical microscope. Near-field evanescent wave information of the sample is transferred into a propagating wave, which reaches the far-field.

A recent effort to overcome this limitation was the introduction of a technique, in which the point of an atomic force microscopy (AFM) tip was replaced with a so-called microsphere superlens [12]. This system could produce super-resolution images by scanning of the AFM head over the sample, but despite of the well-established AFM-positioning control schemes, it required careful consideration of the mechanical system's design, as the detection principle is very sensitive to vibrations. Our goal is to combine a robust scanning system with the super-resolution imaging capabilities of a dielectric microsphere, by which any classical optical microscope can be upgraded to a super-resolution one.

DESIGN AND FABRICATION

The field-of-view of a dielectric microsphere-based super-resolution system cannot exceed a fraction of the size of the microsphere placed on the sample, because the distance an evanescent wave can travel in the optical medium (in this study immersion oil) between the sample and the microsphere before extinction, is extremely small. Our approach was to use this relatively small useful imaging area and translate the microsphere over the full sample to increase the field-of-view. To implement such a scanning mechanism, we fixed 40 μm diameter barium-titanate glass (BTG) microspheres (Cospheric, Santa Barbara, CA, USA) randomly to a 150 μm thin glass slide

using Norland Optical Adhesive 63 (NOA63) optical glue. The height of the NOA63 layer was critical, as imaging was not enabled if the microspheres were covered with the optical adhesive. Then the slide was fixed on a metal frame that was attached to the objective via four metal rods. Before the imaging experiments, it was also needed to adjust the slides horizontally, to position a single BTG microsphere in the field-of-view of the camera that was attached to the optical microscope. We then performed an initial scan to determine the exact position of the virtual image plane. The position, where the biggest magnification and the best contrast were found, was defining the imaging plane. Afterwards an automatic step-by-step scanning and imaging of the sample was carried out. Our acquisition algorithm took a picture at the initial location, then increased the distance between the sample and the microsphere to avoid scratching the sample. Next, it moved one step along the x- or y-axis and set the microsphere in contact for the next image. Later, the generated image set was combined into a single wide-field super-resolution image. To present the super-resolution capabilities of our system, we imaged the surface of a single layer 25 GB capacity Blu-ray disk (without protection film). A Zeiss Axio Imager M2m upright optical microscope equipped with HAL100 halogen light source, an AxioCam MRm camera and a 63× oil immersion objective with a numerical aperture (NA) of 1.4 was used.

RESULTS

Figure 2.A-B show the Blu-ray disk surface as observed by optical microscopy without use of a microsphere and by Scanning Electron Microscopy (SEM), respectively. On the SEM image it is possible to observe both the tracks (with 320 nm pitch distance between them) and the individual data bits (with smallest feature of 150 nm) [13]. Meanwhile on the optical image, even the tracks are not distinguishable. *Figure 2.C* is the image obtained with a BTG microsphere positioned on top of the sample, clearly resolving both the tracks and the individual data patterns on the disk. It is important to note, that the resolution varies across the field-of-view of the microsphere and super-resolution only occurs in a limited area in the center of the microsphere. This area is indicated by the dashed green square in *Figure 2.C*, which we used for reconstructing a composed wide-field image, as shown in *Figure 3*.

As we put the BTG microspheres randomly on the substrate, we observed that, in certain cases, they got stuck to each other (see *Figure 4.A*). This resulted in multiple microspheres within the field-of-view of the microscope objective. Thereby, we realized that it was possible to generate simultaneously super-resolution images through multiple microspheres, as shown on *Figure 4.B*.

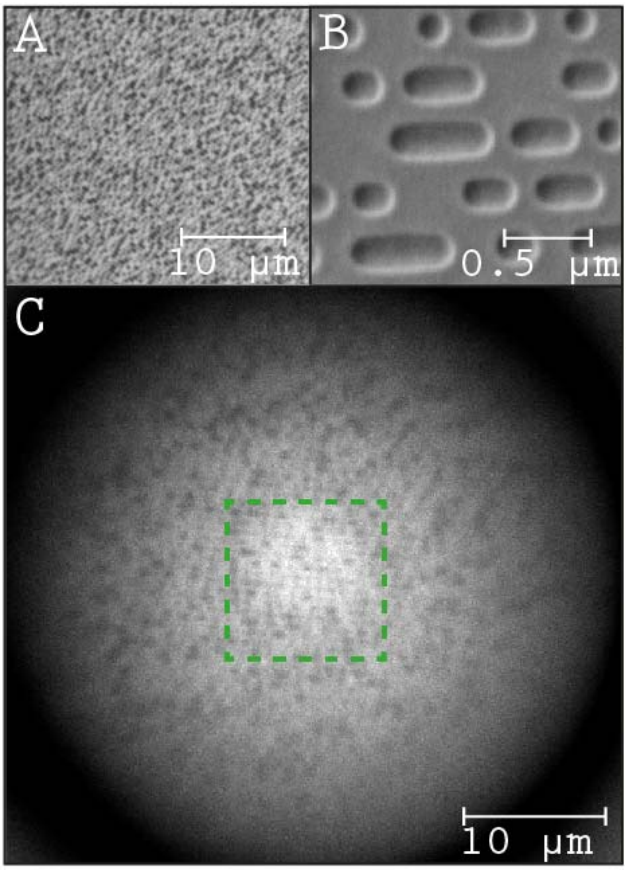

Figure 2: Super-resolution imaging of a Blu-ray disk surface. A: Photograph of the surface of the Blu-ray disk taken without using a microsphere. B: SEM picture of the Blu-ray disk. C: Photograph of a Blu-ray disk surface taken by placing a BTG microsphere on the disk. The dashed green square shows the area underneath the microsphere where super-resolution imaging occurs.

To further exploit this new imaging possibility, we designed a silicon template to generate an array of precisely positioned BTG microspheres. The developed process (see *Figure 5.A*) started with a dry etch step using positive photoresist as mask. This was followed by the back-side grinding of the silicon wafer and the photoresist removal, resulting in a final silicon thickness of 150 µm for formation of the through-hole template. Then BTG particles were dissolved in deionized water and placed on the top of the wells in a 6 µL droplet that was swiped over the surface a couple of times (*Figure 5.B*). With this method it was possible to guide the microspheres over the fabricated through-holes of the template. As the filling rate reached sufficient level, which was easily achievable, since only a 2×2 microsphere array could be positioned into the field-of-view of the 63× objective, it was crucial to remove the excess of the microspheres, otherwise they would act as

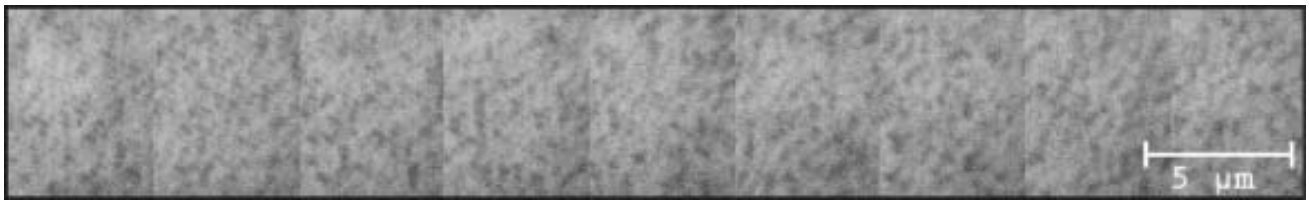

Figure 3: Larger field-of-view image of a Blu-ray disk surface composed of individual photographs taken during scanning with a BTG microsphere immersed in oil.

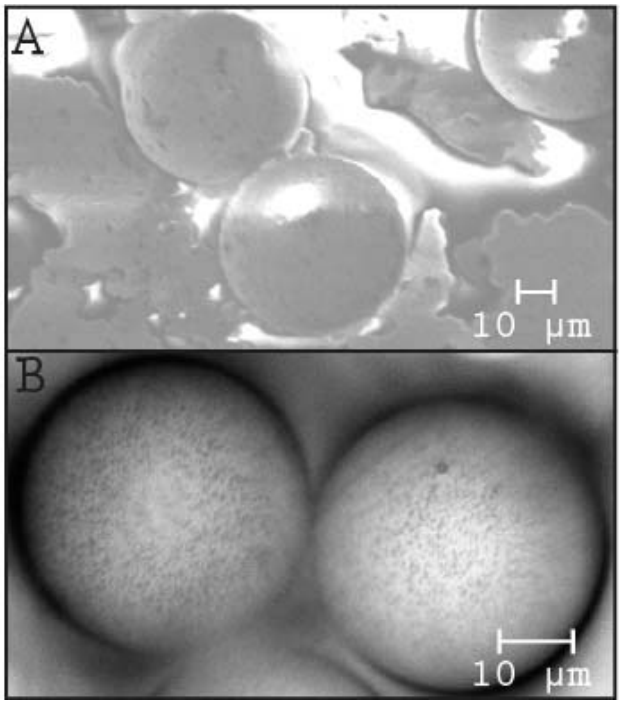

Figure 4: Parallelized imaging with randomly positioned BTG microspheres. A: SEM photograph of an ensemble of randomly positioned microspheres attached to a 150 µm thin glass slide with NOA63 optical glue. B: Image of the Blu-ray disk surface taken via two microspheres.

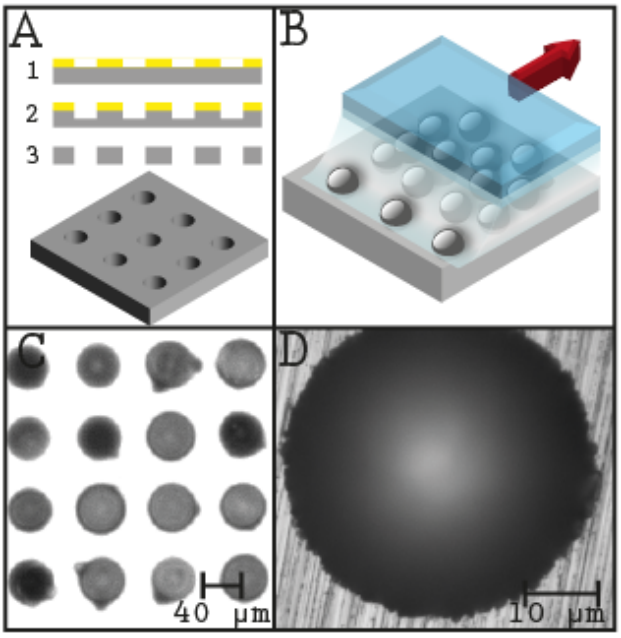

Figure 5: Microsphere array fabrication. A: Fabrication process of a through-hole silicon template and schematic of the final chip, with (1) patterning of the photoresist, (2) dry etching of holes using the Bosch process, and (3) wafer grinding resulting in through-holes and resist removal. B: Schematic of the patterning process of the BTG microspheres on the silicon template. Optical photographs of C: a chip with BTG microspheres positioned on top, and D: a zoom from the back on a single hole on which a BTG microsphere is positioned.

a spacer between the sample and the silicon substrate. Imaging experiments will be performed in the framework of an extensive future study of the capabilities of this new multi-microsphere system.

CONCLUSIONS

We showed how a simple but smart dielectric microsphere-based optical system can be constructed. Our design consisted of BTG microspheres, which can be used for super-resolution imaging if immersed in oil. We showed imaging capability of our setup by imaging the surface of a Blu-ray disk. We also presented a stitching process, consisting of an automated microscope stage control, combined with image acquisition and post-processing algorithms. Thereby, we managed to overcome the field-of-view limitation of a single microsphere. We realized the possibility of multi-microsphere imaging by designing a silicon-based template to create an ordered array of microspheres, which could lead to four times faster imaging, considering the field-of-view limitations of our microscope mounted camera.

Our system allows super-resolution imaging in an affordable way and just requires a classical optical microscope. We therefore think that our findings may be at the basis of a future generalized and affordable scanning super-resolution optical microscopy with huge potential in many research fields.

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COMPETING FINANCIAL INTEREST

The authors have an International Patent Application n° PCT/IB2014/063747, filed on August 6, 2014, for a device that is related to this work.

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